How reliable is the dxFeed demo feed?

Data consistency, behaviour under load, and what I would do about it

Irakli · L2 Support Engineer assessment · Task C

Measured on Friday 16 October 2026 · every number reproducible offline from evidence/

CONTEXT

What I will cover

01	**Task A calendar**	Where the holiday data comes from, and what I could not get	3 min
02	**Data consistency**	Delay, the silent history cap, volume, prices, FX sessions	15 min
03	**Behaviour under load**	What happens when I add parallel requests	9 min
04	**Risk, incident, improvements**	What I would escalate, a drill, three proposals	10 min
	Questions		5 min

CONTEXT

The short version

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58,315

candles pulled and checked

900 s

delay on every symbol: the feed is 15 min behind

8,000

candles per series, then it stops and still says OK

130 + 1

AAPL hours where 1h volume ≠ sum of 5m (ETH + RTH)

50

EUR/USD 1-minute candles outside the FX session

~3 req/s

throughput ceiling; more workers just wait longer

0

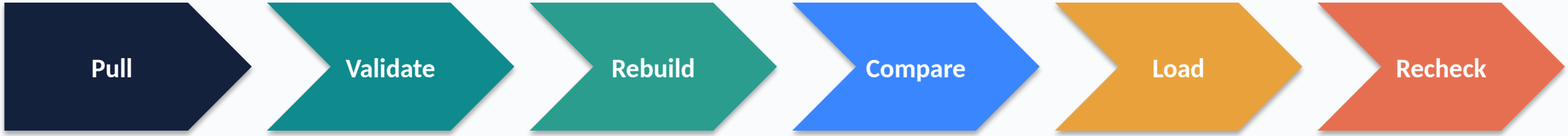
HTTP errors in 648 load requests, and no 429

60

holiday rows from the Task A scraper

CONTEXT

How I worked



Pull	Validate	Rebuild	Compare	Load	Recheck
96 chunked requests, 4 instruments, 1m / 5m / 1h	gaps, order, grid, NaN, sessions, holidays, DST	5m from 1m, 1h from 5m, compare candle by candle	SSE stream and the Radar web tool	1 → 50 workers with automatic stop rules	every headline number recomputed from disk

Instruments

AAPL · US stock

TSLA · high volatility (no crypto on demo)

EUR/USD · FX, bid candles only

/ESZ26 · futures, used as a control

Logged for every call

URL and parameters, UTC time

latency, status, body hash

raw body saved (gzip)

Polite by design

one client, think time 0.25 s

stop on 429, errors, p50 > 8 s

budget of 3,500 requests

Rule I kept: a number goes on a slide only if I can recompute it offline.

01

Task A: holiday calendar

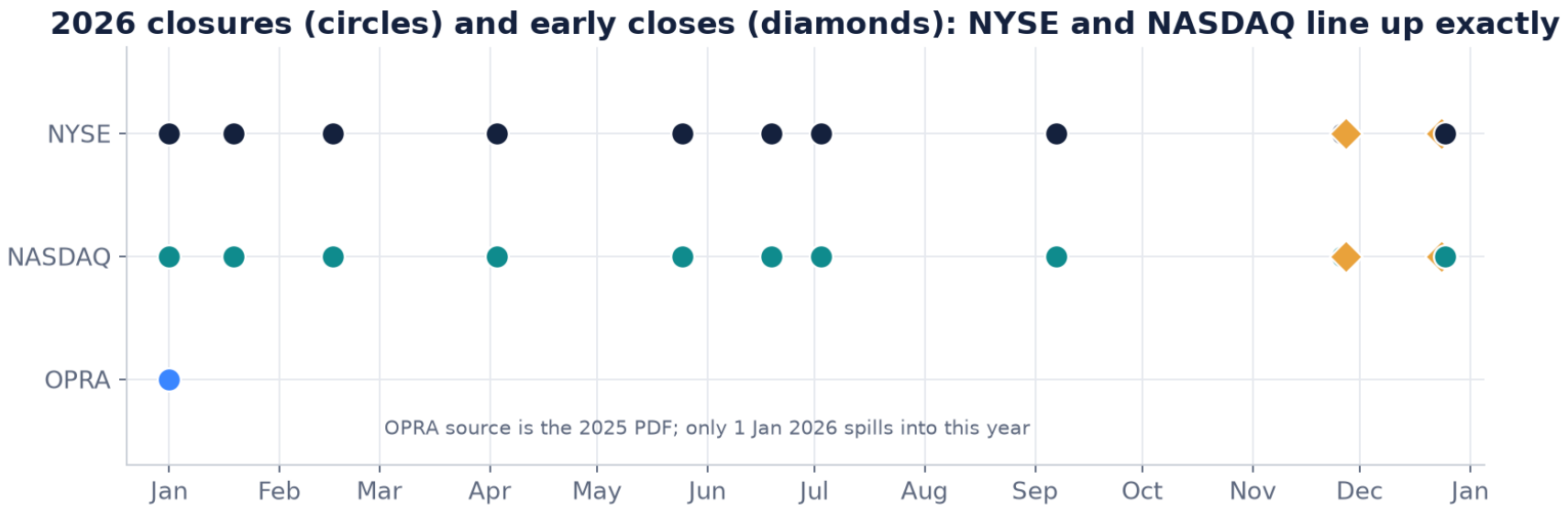
The calendar Task C uses to tell a holiday from a gap

about 3 minutes

TASK A CALENDAR

NYSE and NASDAQ agree on every 2026 date

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1. **60 rows:** NYSE 34, NASDAQ 12, OPRA 14.
2. **NASDAQ** publishes the current year only.
3. **OPRA** 2026 PDF not out yet (403).
4. **Drive:** uploaded by hand; API path untested.

60 rows, weekly on GitHub Actions. CME is not scraped because its terms forbid it.

02

Data consistency

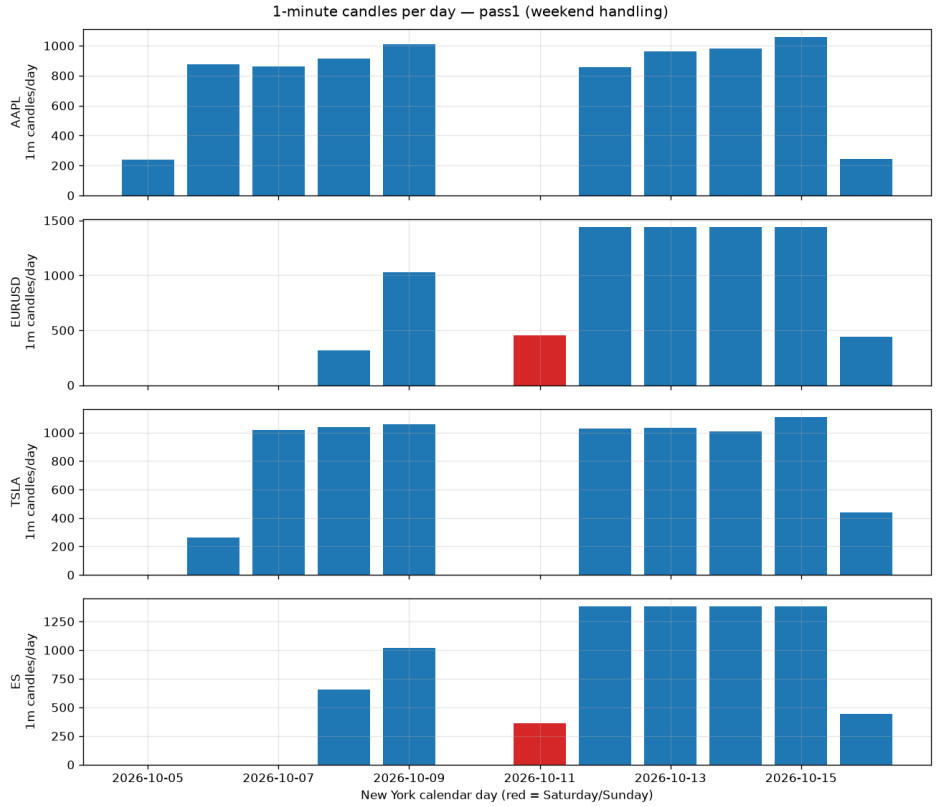
What is right, and the problems that do not raise an error

about 15 minutes

DATA CONSISTENCY

What is right: the basics are clean

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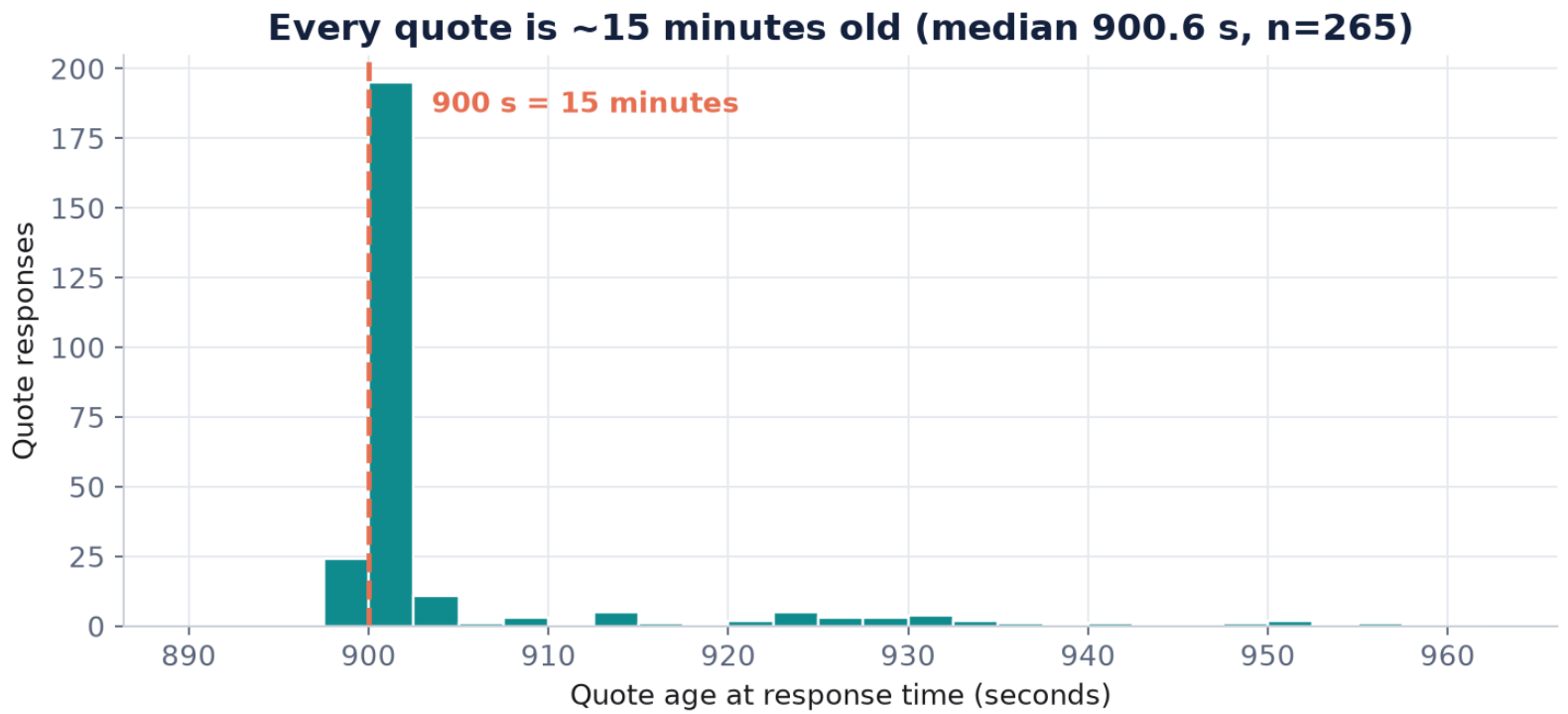
1. **0** timestamps off the grid, out of order, or NaN.
2. **Holidays** match my Task A calendar exactly.
3. **21 duplicates** at chunk edges, all identical.
4. **DST** handled; first bar is 00:00 ET.

58,315 candles: the price path is sound. The problems are quieter than that.

DATA CONSISTENCY

Every quote is exactly 15 minutes old

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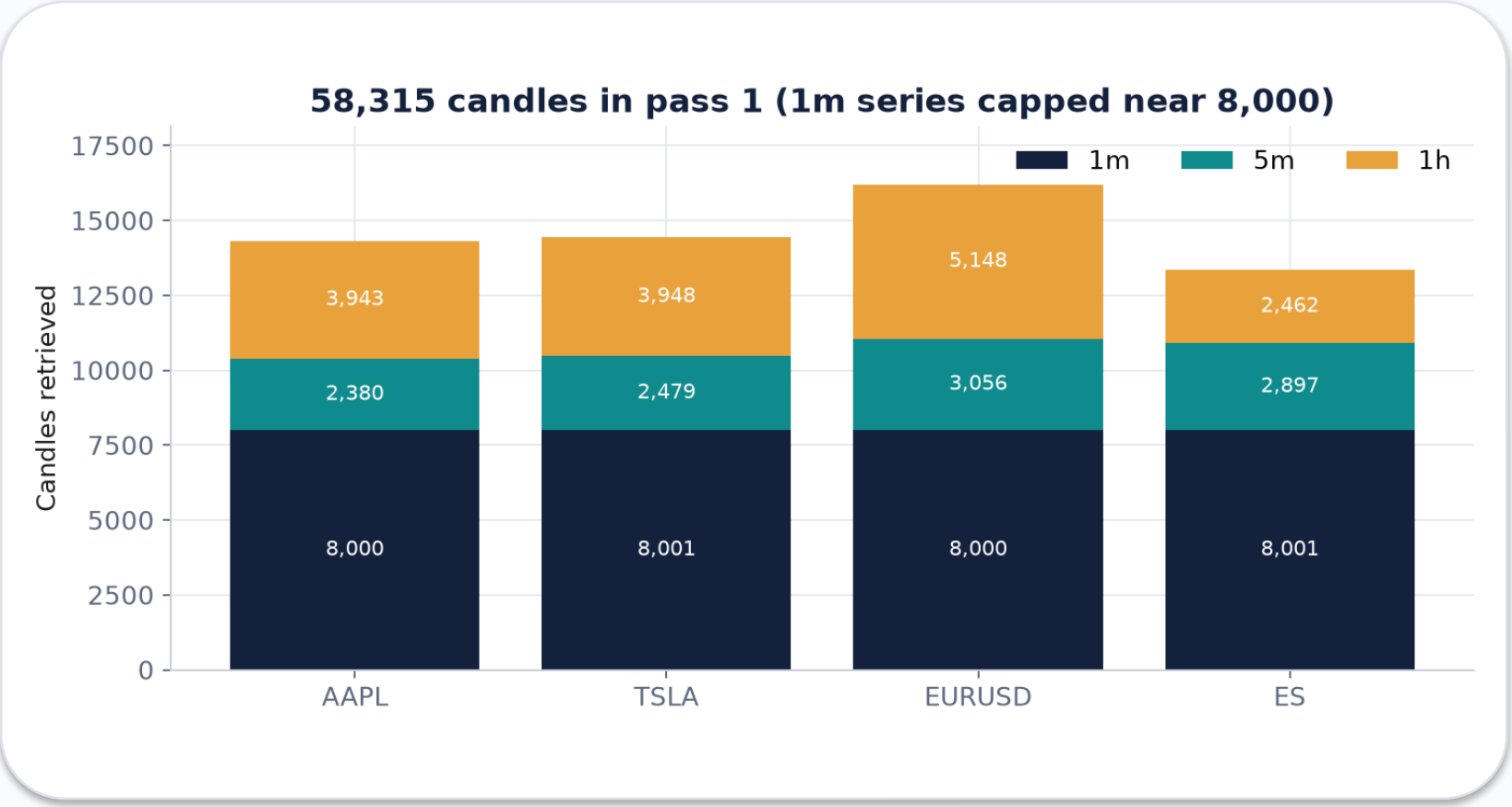
1. **Median 900.6 s** on REST, SSE and Radar.
2. **Load** did not change it. Latency moved, freshness did not.
3. **Nothing** in the payload says the data is delayed.

Any freshness alert has to start from a 900-second floor, not from zero.

DATA CONSISTENCY

History stops at 8,000 candles and still says OK

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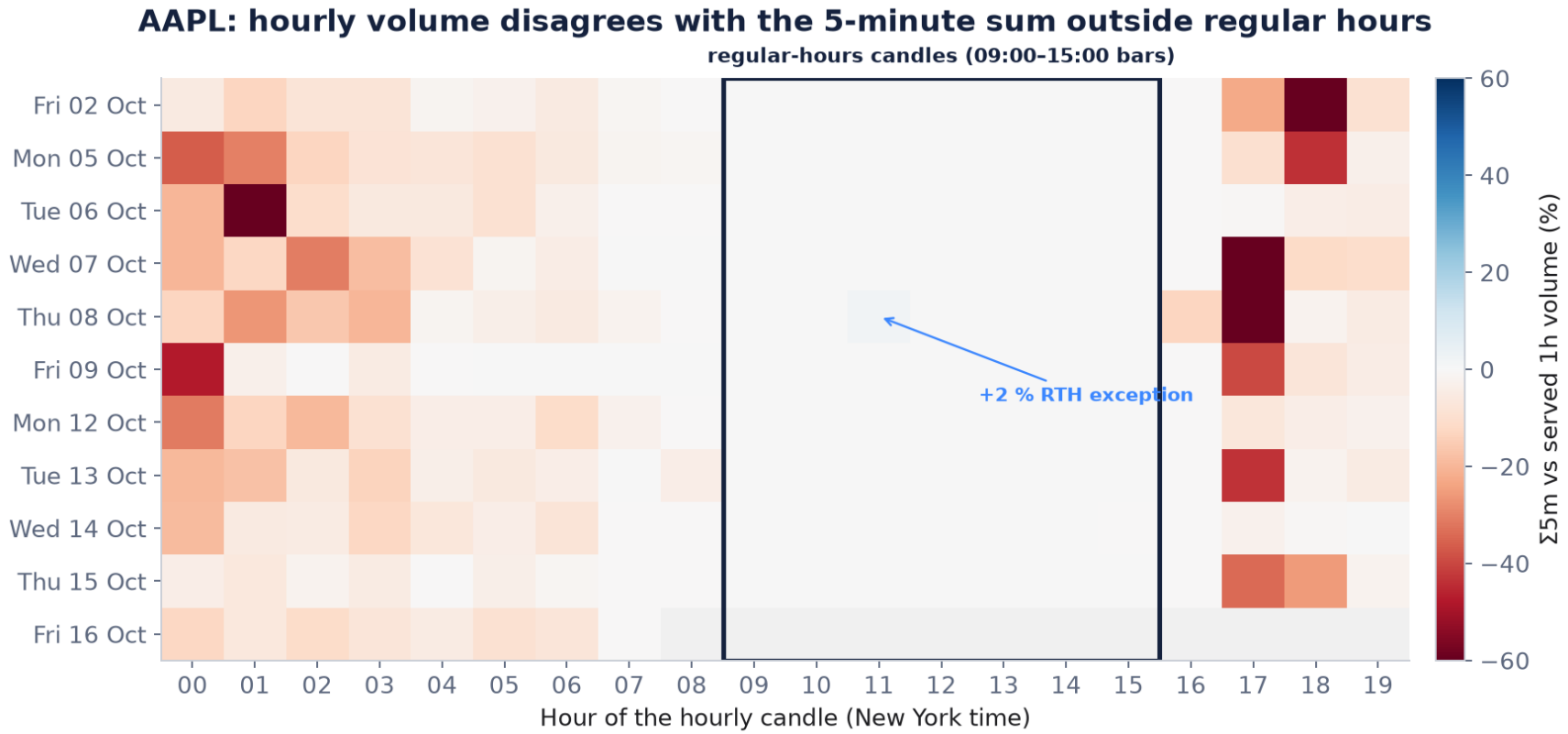
1. **All four** 1-minute series stop at 8,000 or 8,001.
2. **Older window:** status OK, empty body, 15 bytes.
3. **So it is retention,** not a response-size limit.

A two-week backtest silently loses a week. To the client it looks like a holiday.

DATA CONSISTENCY

Hourly volume breaks outside regular hours

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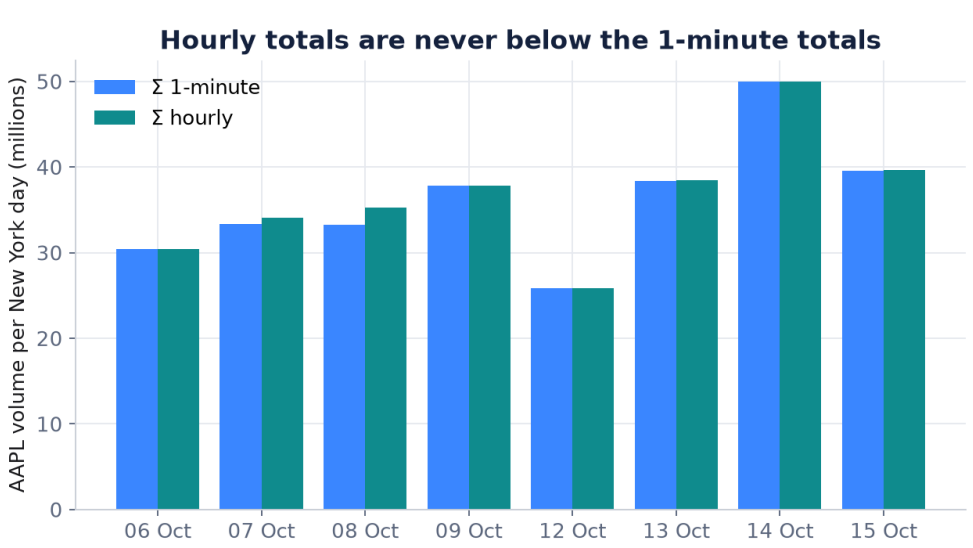
1. **Red** = hourly bar has more volume, up to 99 %.
2. **Boxed** regular hours are clean except one cell.
3. **TSLA** shows the same shape.
4. **1m and 5m** agree on every complete bucket.

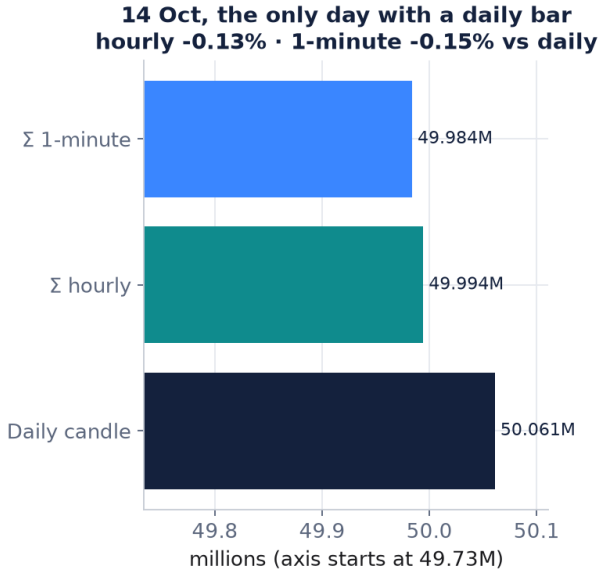
AAPL: 130 extended-hours mismatches and 1 in regular hours. TSLA: 118 + 2. Cause not proven.

DATA CONSISTENCY

Which volume is right? Not decided yet

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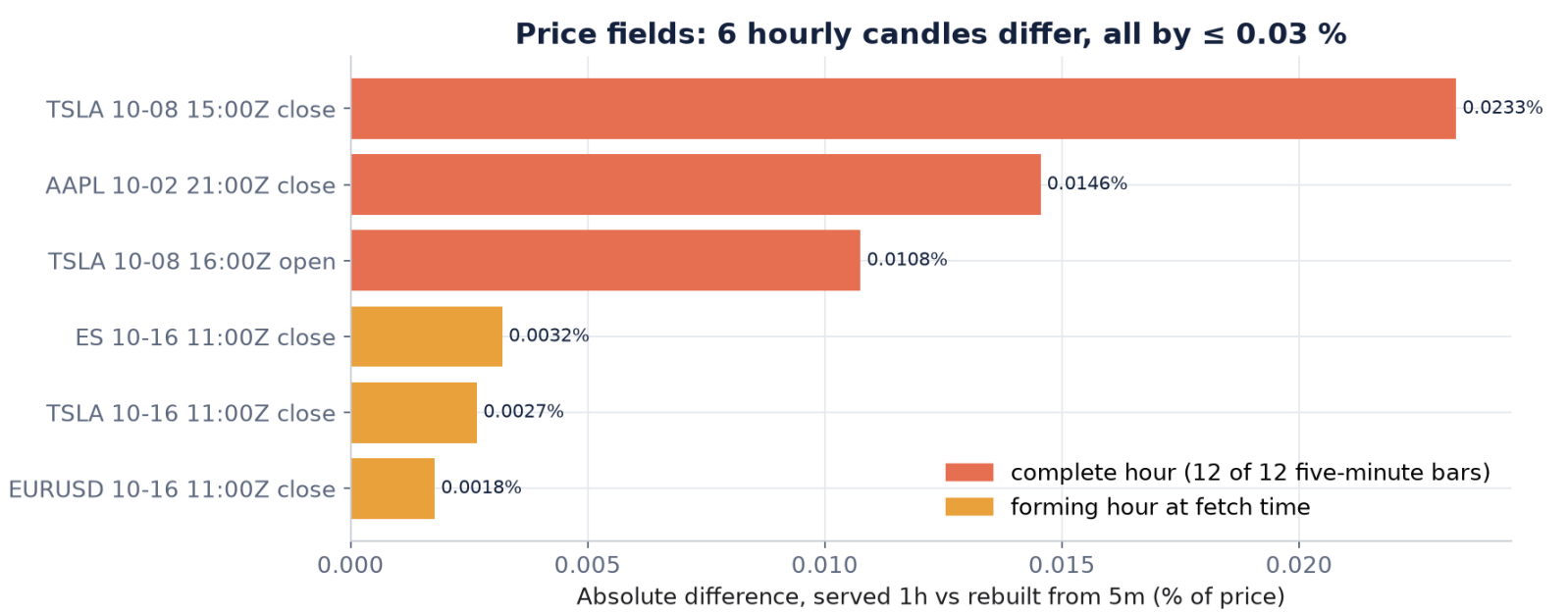
1. **14 Oct daily candle:** hourly −0.13 %, 1-minute −0.15 %.
2. **Hourly** is never below 1-minute on any day.
3. **One day** is not enough to pick a reference.

What I would tell a client: we are confirming which series is the reference. No workaround yet.

DATA CONSISTENCY

Prices are almost, not perfectly, consistent

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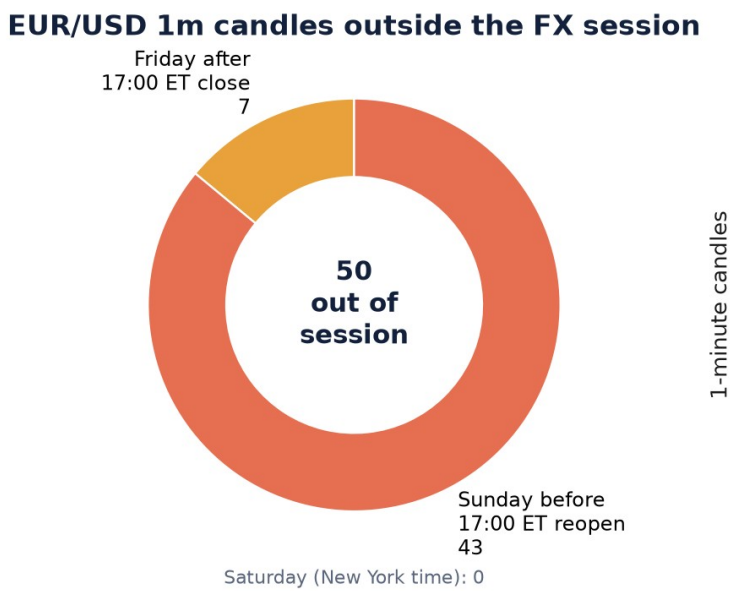
1. **6 hourly candles** differ from their 5-minute rebuild.
2. **3** are the bar still forming at fetch time.
3. **3** are complete hours. Largest gap 0.023 %.

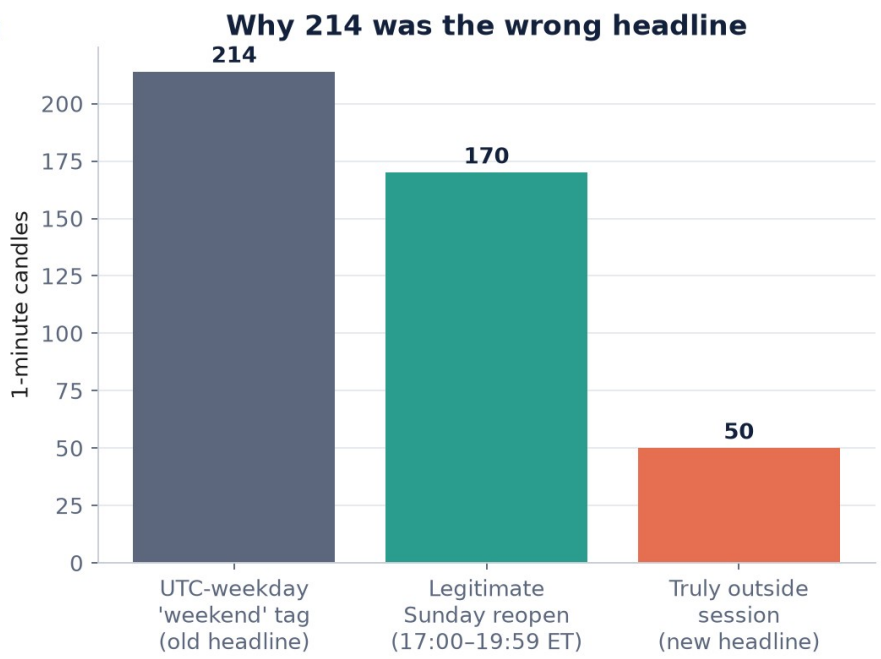
I first wrote 'OHLC always agrees'. It doesn't. The gaps are tiny, but real.

DATA CONSISTENCY

EUR/USD prints 50 candles while FX is closed

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1. **7** after the Friday close, 0 Saturday, 43 Sunday early.
2. **170** Sunday-evening candles are the normal reopen.
3. **ES futures:** 0 outside session, so the method works.

I first counted 214 by UTC weekday. Against the real FX session it is 50.

DATA CONSISTENCY

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Small things that turn into tickets

Where	What happens	What the client sees
EUR/USD{=1m}	OK, no candles. Only price=bid works	Silent empty result
Daily candle	Stamped 00:00 UTC = 20:00 ET the day before	Wrong date once shown in ET
toTime	Inclusive: 60 minutes returns 61 candles	Duplicates at chunk edges
Unknown symbol	HTTP 200 + TIMED_OUT after 3.3 s (29 s under load)	Looks like success, slows everything
Forming candle	No flag on it (eventFlags = 0)	Client stores a bar that changes
SSE in delay window	NaN candles dated 2038-01-19	Parsers store garbage
Radar vs REST	Radar shows NVDA, REST times it out	'But it works in Radar'

None of these is an error. Each one cost me an experiment, so each will cost a client a ticket.

03

Behaviour under load

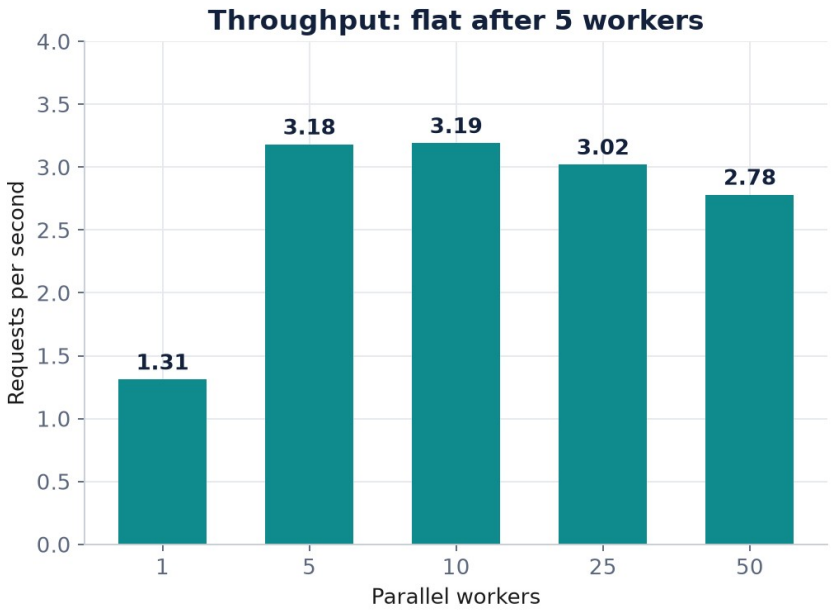
What happens when one client adds parallel requests

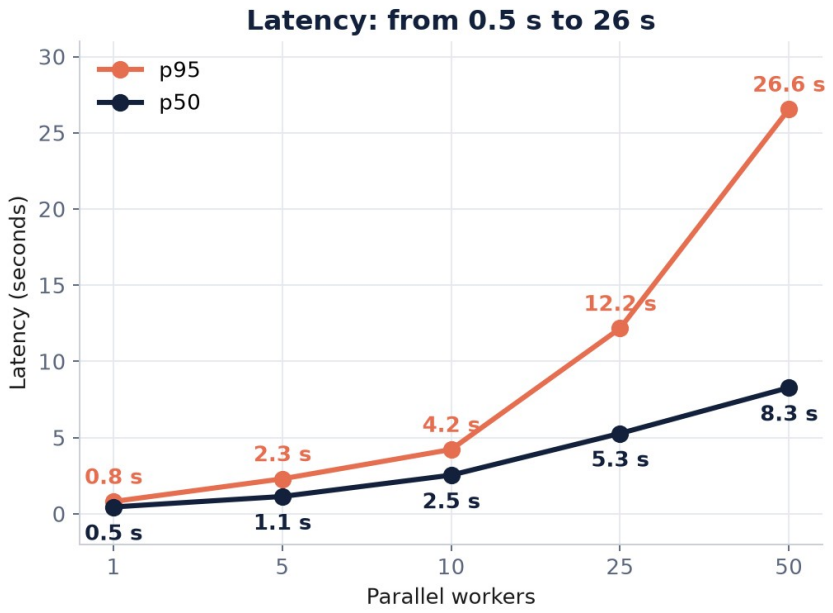
about 9 minutes

BEHAVIOUR UNDER LOAD

More workers did not buy more throughput

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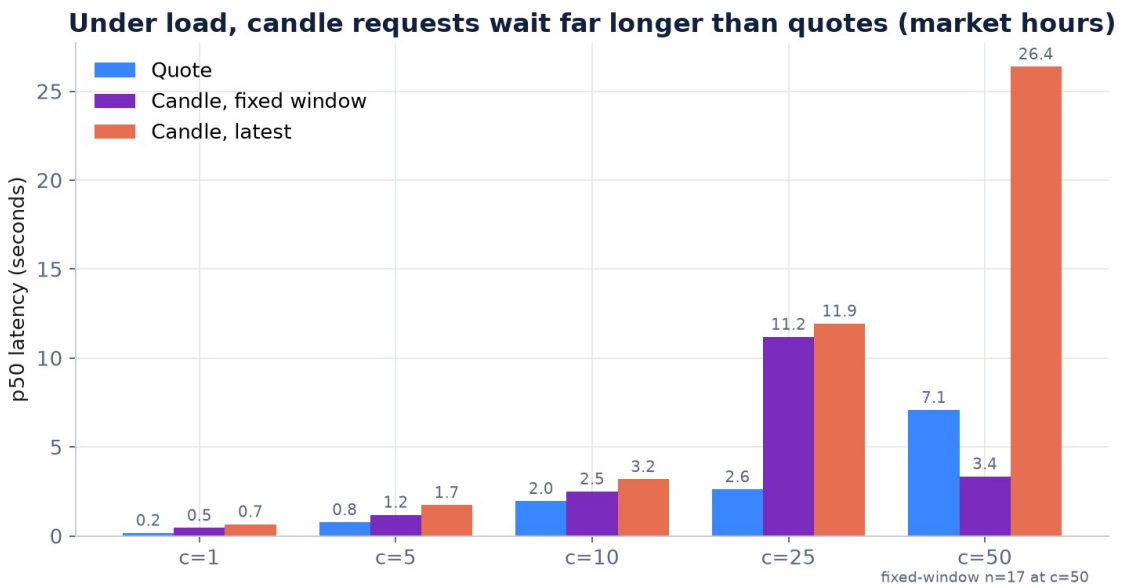
1. **Pre-market** plateau was lower: ~1.6 req/s.
2. **648** requests, 0 HTTP errors, no 429.
3. **Earlier '4.6 req/s'** ignored in-flight requests.

Throughput stays near 3 req/s while median latency goes 0.5 s → 8.3 s. Seconds, not milliseconds.

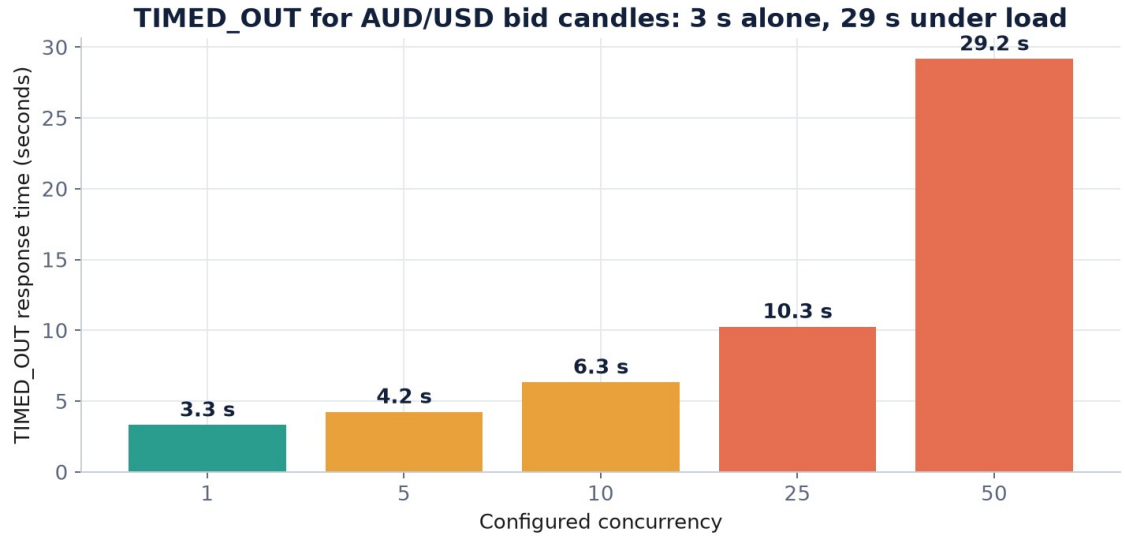
BEHAVIOUR UNDER LOAD

Candle requests queue first, and timeouts stretch

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Median latency by request type (seconds)



TIMED_OUT response time (seconds)

At 25 workers quotes take 2.6 s, candles 11–12 s. An unknown symbol goes from 3.3 s to 29 s.

Percentiles count completed requests only, so at 25 and 50 workers the real tail is worse.

BEHAVIOUR UNDER LOAD

Was the bottleneck my client? No.

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in-flight = setting

Little's law at 5 and 10 workers: my workers really were all waiting on the server.

52 ≥ 50

Connection pool limit was above the highest concurrency, so it could never bind.

1 client

Per-IP throttle or shared capacity? I can't tell without a second IP. Not run.

This shows how this shared demo behaves for one client. It is not a production capacity number.

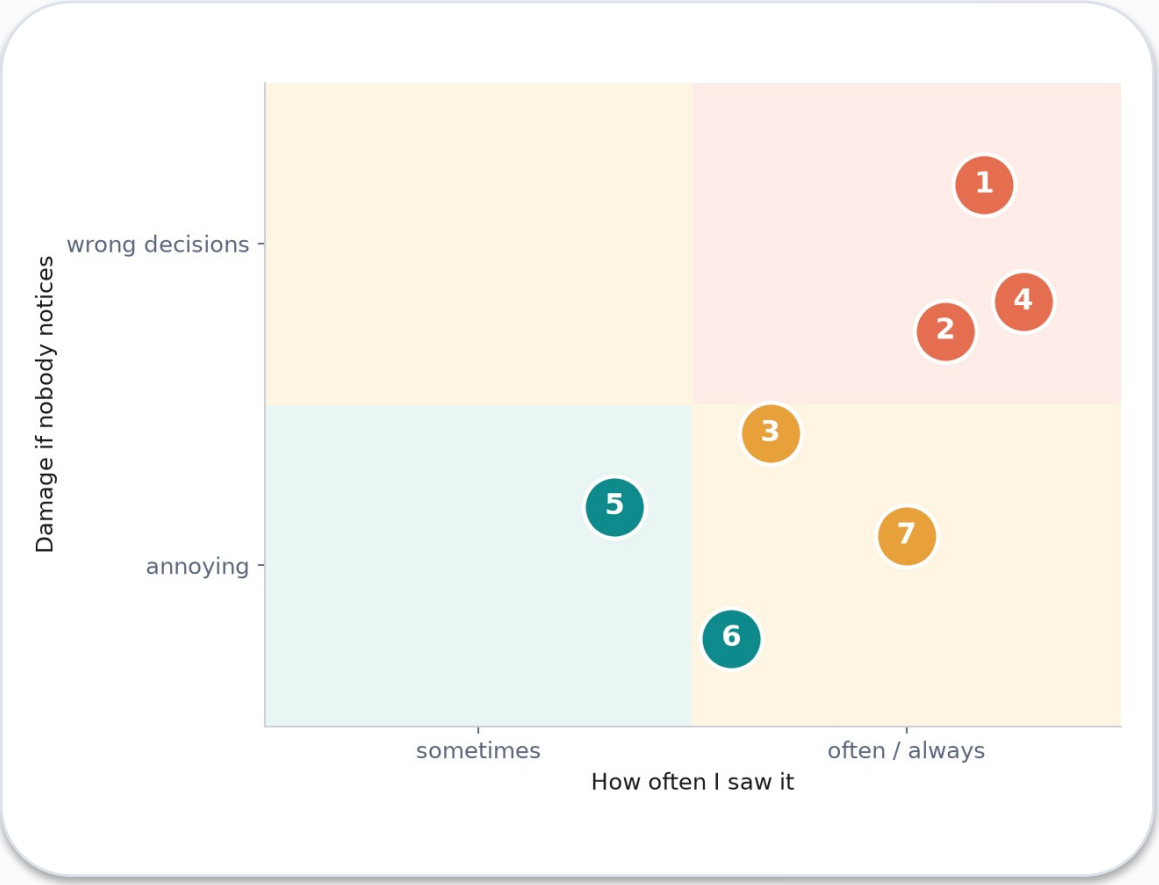
04

Risk, incident and improvements

What I would escalate, how I'd run the incident, what I'd fix

about 10 minutes

What I would escalate, in order

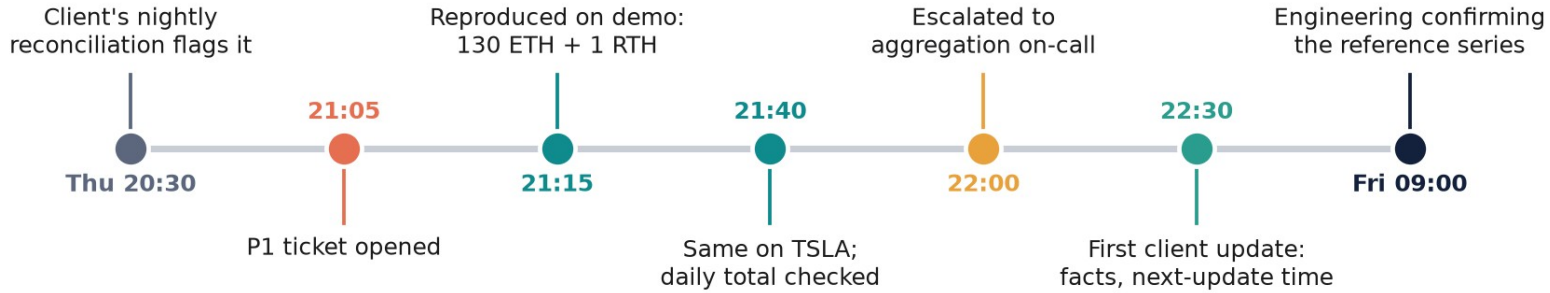


1. **Silent 8,000 cap**
2. **Volume depends on timeframe**
3. Queues, never says 429
4. Stale looks like closed
5. Symbol quirks (TIMED_OUT, FX bid)
6. FX prints outside session
7. Monitors only watch HTTP errors

Silent beats loud: a wrong answer with status OK is worse than an error.

SIMULATION

Incident drill: 'AAPL hourly volume is corrupted'



First 15 minutes

Reproduce from public endpoints: −98.4 % on 2 Oct 18:00 ET.

Count: 130 extended-hours, 1 regular-hours.

Confirm 5m = 1m; check the daily total.

Escalate with the CSV and one question: which series is the reference?

What the client hears

“We reproduced it. It affects hourly volume for US stocks, almost only outside regular hours. Prices agree. We are confirming the reference series. Next update 10:00 UTC.”

No root cause, no untested workaround.

Three things I would change

PROPOSAL

TECHNICAL

Say what the response is

- Add truncated, serverTime, delaySeconds to the response.
- Flag the forming candle.
- Return UNKNOWN_SYMBOL at once, not TIMED_OUT.

Effort **Medium**

Impact **Removes the top two 'wrong data' tickets**

MONITORING

Catch it before the client does

- One synthetic probe per asset class.
- Age vs calendar, 1h vs 1m volume, p95 at low concurrency.
- Budget the probes: the service queues.

Effort **Small–medium**

Impact **Client-found incidents become ours**

DOCUMENTATION

One 'demo facts' page

- Delay, retention cap, symbol list.
- FX needs price=bid; toTime is inclusive.
- Radar vs REST differences.

Effort **Small**

Impact **Fewer first-contact tickets**

Four things I got wrong first, and fixed

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"214 weekend FX candles"	50 outside the real FX session	I counted by UTC weekday
"Throughput rises to 4.6 req/s"	Flat at ~3 req/s	I ignored requests still in flight
"OHLC always agrees"	6 small differences, ≤ 0.023 %	My first check was too loose
"Rebuild ETH volume from 1m"	Reference series not decided	The daily candle says otherwise

Re-running everything offline is what caught these. That re-check is now part of the method.

What is still open

NOT DONE

Waiting on the clock

- Pass 2 re-pull after Monday 19 Oct 15:30Z
- Weekend logger: stopped Sat 14:28Z, Sunday not covered

Blocked on the demo

- TimeAndSale / Trade history: TIMED_OUT or empty
- So the volume cause stays a hypothesis

Not run

- SPY / MSFT check of 8 Oct 11:00 ET; tho=true
- Second client / IP; burst and batching tests

Task A

- CME: manual files only (terms)
- OPRA 2026 PDF not published; Drive API untested

Questions

Happy to open any evidence file live.

I can show live

The 8,000 cap: a long fromTime, or chunks.jsonl

The AAPL −99 % and +2 % volume rows in the CSV

Radar screenshots and the SSE comparison

I won't claim

A root cause for the volume split

That the FX prints are trades

A production capacity number